COMMUNICATION: The act of sending information from one place to another is called communication. The information could be of various types such as text, sound, video, kinesthetic (involving physical movement) or a combination thereof. Communication plays a very vital role in our civilization. It forms the basis of collaboration and coordination of various useful projects as well as it is required for entertainment and awareness.

MAJOR MILESTONES IN HISTORY OF COMMUNICATION SYSTEMS:

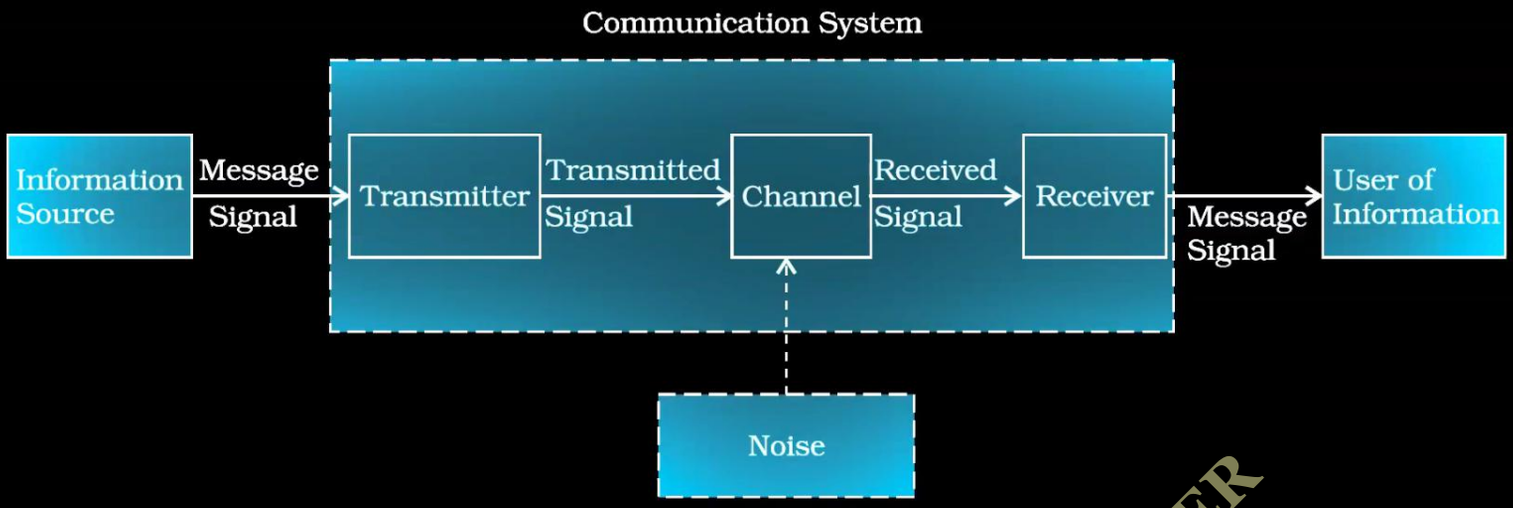
TABLE 15.1 SOME MAJOR MILESTONES IN THE HISTORY OF COMMUNICATION

Year	Event	Remarks
Around 1565 A.D.	The reporting of the delivery of a child by queen using drum beats from a distant place to King Akbar.	It is believed that minister Birbal experimented with the arrangement to decide the number of drummers posted between the place where the queen stayed and the place where the king stayed.
1835	Invention of telegraph by Samuel F.B. Morse and Sir Charles Wheatstone	It resulted in tremendous growth of messages through post offices and reduced physical travel of messengers considerably.
1876	Telephone invented by Alexander Graham Bell and Antonio Meucci	Perhaps the most widely used means of communication in the history of mankind.
1895	Jagadis Chandra Bose and Guglielmo Marconi demonstrated wireless telegraphy.	It meant a giant leap – from an era of communication using wires to communicating without using wires. (wireless)
1936	Television broadcast(John Logi Baird)	First television broadcast by BBC
1955	First radio FAX transmitted across continent.(Alexander Bain)	The idea of FAX transmission was patented by Alexander Bain in 1843.
1968	ARPANET- the first internet came into existence(J.C.R. Licklider)	ARPANET was a project undertaken by the U.S. defence department. It allowed file transfer from one computer to another connected to the network.
1975	Fiber optics developed at Bell Laboratories	Fiber optical systems are superior and more economical compared to traditional communication systems.
1989-91	Tim Berners-Lee invented the **World Wide Web**.	WWW may be regarded as the mammoth encyclopedia of knowledge accessible to everyone round the clock throughout the year.

YOUTUBE.COM/@THEPHYSICSSHELTER

AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

Block Diagram Of A Generalized Communication System:



Block diagram of a generalised communication system.

The above chart represents the generalised block diagram of communication system. Every communication system has three essential elements namely:

Transmitter: It transmits the information in to the channel or medium which carries this information to a different point in space. For instance, in radio stations, transmitter emits signals in the form of electromagnetic waves which are carried through atmosphere to various places. Apart from transmitting, the transmitter also processes the raw message signal recieved from the source to make it suitable for transmission.

Medium /Channel: The physical entity that carries the transmitted signal from one point of space to another is called medium or channel. A channel could either be wireless (such as surrounding atmosphere which may carry radio signals) or wired (such as broadband DSL networks or telephone landlines).

Reciever: The device which receives the signal from the channel is called reciever. The reciever processes the recieved signal and sends it to the user of the information. This can require

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

reconstruction and decoding of recieved signal in to a recognisable form for the information users.

TRANSDUCER: If the output of the information source is non electrical (for instance voice, temperature or pressure etc), the transducer converts it in to electrical form before giving it as an input to the transmitter. For instance for voice signals, a microphone can act as a transducer. Similarly we can have other types of transducers to convert temperature, pressure, velocity etc signals in to electrical form.

NOISE AND DISTORTION: When a signal propagates through the medium, it might become distorted due to properties of the channel itself. This is known as distortion of signal. Besides distortion signal can also become corrupted due to mixing of unwanted electrical disturbances from random sources. The mixing of unwanted electrical disturbances on the message signal is called noise. It leads to a corrupted version of the signal.

BASIC MODES OF COMMUNICATION: There are two basic modes of communication:

POINT TO POINT COMMUNICATION: Here the communication takes place between a single transmitter and a single recievers. For example, Telephony, where in one person can talk with another person (only two telephones are involved).

BROADCAST: In broadcast mode, there are large number of recievers corresponding to a single transmitter. Radio, Television and live streaming (such as on youtube etc) are examples of broadcast.

YOUTUBE.COM/@THEPHYSICSSHELTER

Basic Terminology Of Communication Systems:

Transducer: Any device which converts one form of energy into another is called a transducer. For electronic communications, we generally require transducers which have either their inputs or outputs in electrical form (for example output of a microphone and input of a speaker).

Electrical Transducer: A device which converts some physical variable (pressure, displacement, force, temperature etc) into corresponding variations in electrical signals at the output is called an electrical transducer.

Signal: Information converted to electrical form suitable for transmission is called signal. Signals can be classified as either Analog signals or digital signals as will be described below.

Analog Signal: Analog signals are continuous variations of voltage or current. They are essentially single valued functions of time. Sine wave is the fundamental analog signal. All other analog signals are usually analysed by considering them as superposition of sine waves. Thus analog signals can be thought of as made up of several sine wave components.

Digital Signals: The signals that can take only discrete step-wise values are called digital signals. In digital electronics, the digital signals are interpreted in terms of binary values of signals. Typically a low level of voltage or current is assigned a value 0 and a high value of voltage or current is assigned a value 1. There are certain coding schemes based on binary number system to represent various letters and numbers. For example In a binary coded decimal (BCD) system, each digit is represented by four binary (0 or 1) bits for instance the numbers 0, 1, 2, 3 and 4 are written as 0000, 0001, 0010, 0011 and 0100. 1000 would represent 8. Similarly the ASCII (American standard

YOUTUBE.COM/@THEPHYSICSSHELTER

code for information exchange) is a character encoding which uses numbers to encode letters and various characters.

NOISE: Unwanted signals that tend to disturb transmission and processing of message signals in a communication system are termed as noise. A noise source could be located either inside or outside the communication system.

TRANSMITTER: A transmitter processes incoming message signal so as to make it suitable for transmission through a channel and subsequent reception.

RECIEVER: A recieven extracts the desired message signals from the recieved signals at the output of the channel.

ATTENUATION: Loss of signal strength while propagating through a medium is called attenuation.

AMPLIFICATION: The process of increasing the amplitude (i.e strength of the signal) is called amplification. This is done using an amplifier such as the one we studied in the chapter on semi-conductors. The energy for amplification comes from DC sources. Amplification is done at a place between source and destination whenever the signal strength becomes weaker than the required value.

RANGE: Largest distance between a source and destination at which signal is recieved with sufficient strength is called range.

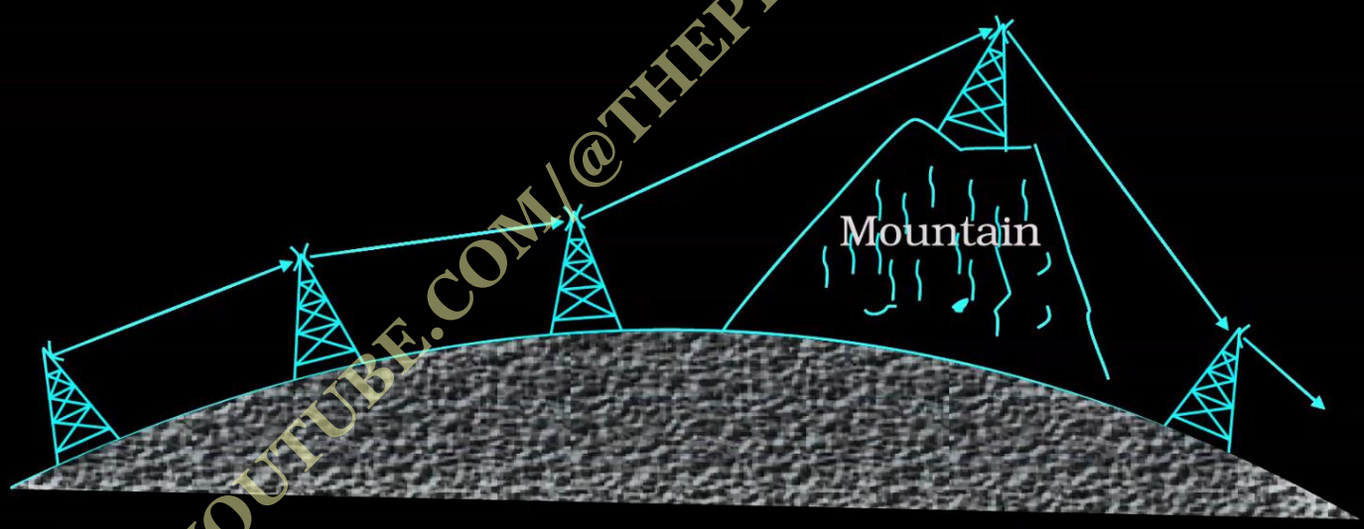
BANDWIDTH: For a communication equipment, bandwidth refers to the range of frequency which can be processed by the equipment while for signals, it refers to the range of frequencies (or the portion of electromagnetic spectrum) occupied by a signal.

YOUTUBE.COM/@THEPHYSICSSHELTER

MODULATION: The original low frequency message / information cannot be transmitted to long distances due to reasons which will be discussed in a later section. For long distance transmission, the low frequency message signal is superimposed on a high frequency waves (known as carrier waves) in various ways. This process is known as modulation. There are several types of modulation which we will describe in more detail in later sections (such as Amplitude modulation (AM), frequency modulation (FM) and phase modulation (PM) etc).

DEMODULATION: The process of retrieval of information from the carrier wave at the receiver is called demodulation. (This is the reverse process of modulation).

REPEATER:



Use of repeater station to increase the range of communication.

A repeater is a combination of a receiver and a transmitter. A repeater picks up signals from a transmitter and then amplifies it before retransmitting it (sometimes with a change in carrier frequency). (communication satellites are also essentially repeater stations in space.

YOUTUBE.COM/@THEPHYSICSSHELTER

BANDWIDTH OF SIGNALS: As described earlier, communication signals can be thought of as superposition of several sinusoidal waves of different frequencies. The range of frequencies forming a signal is called the bandwidth of the signal. Generally with increase in bandwidth, the signal processing becomes more expensive and sophisticated, therefore bandwidths of various types of signals are kept limited according to desired quality of output. The essential bandwidths for some types of signals are described below.

SPEECH SIGNALS: For speech signals frequency range of 300 Hz to 3100 Hz is considered adequate accordingly the bandwidth of speech signals is 2800 Hz (i.e 3100 Hz − 300 Hz), for the purpose of commercial telephonic communication.

MUSIC SIGNALS: Human beings can hear sound frequencies from 20Hz to 20 kHz. Musical instruments produce frequencies over this range and accordingly for satisfactory transmission of music we need a bandwidth of 20 kHz (≈ 20 kHz − 20 Hz).

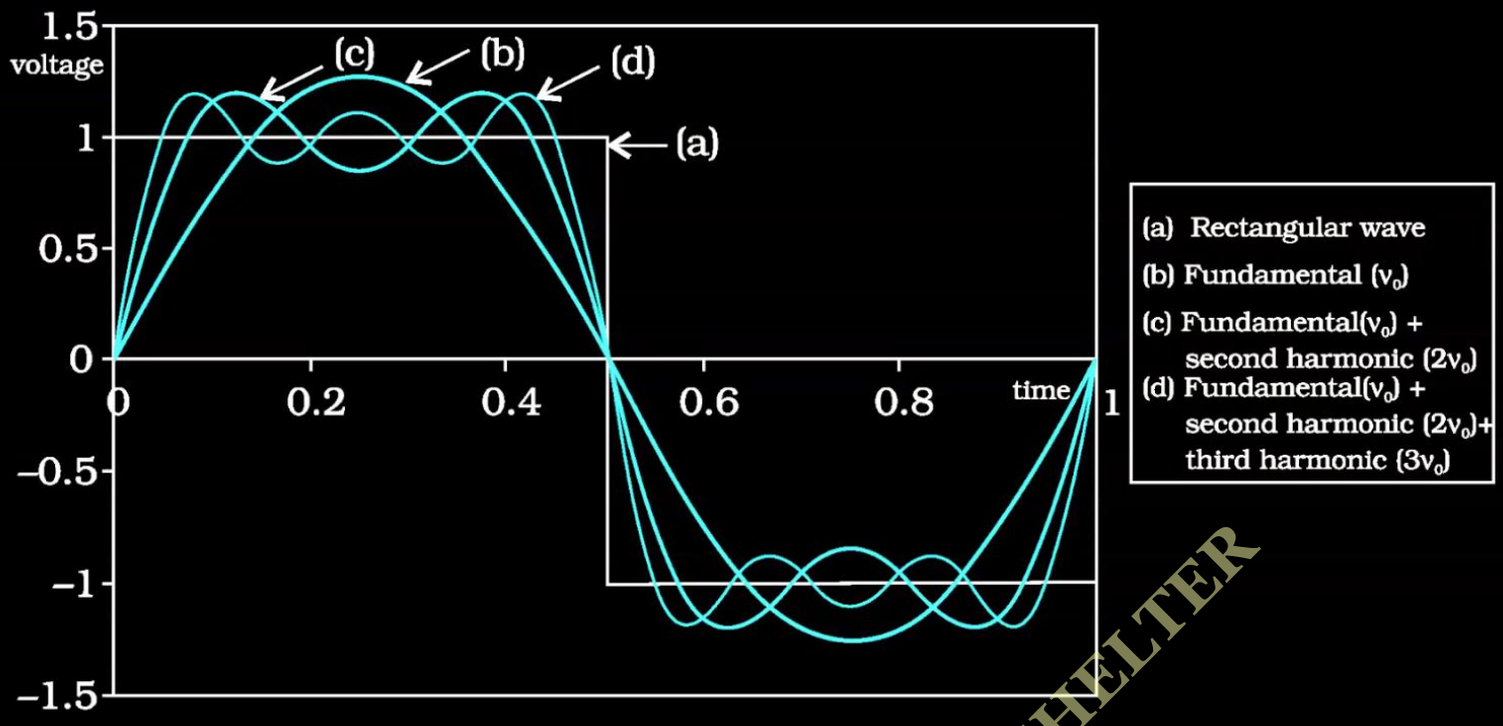
VIDEO SIGNALS: Typical transmission of video signals requires 4.2 MHz of bandwidth.

TV SIGNALS (AUDIO VISUAL): Television signals simultaneously contain audio and video signals and typically require 6 MHz of bandwidth.

YOUTUBE.COM/@THEPHYSICSSHELTER

GENERATION OF DIGITAL SIGNALS BY SUPERPOSITION OF SINUSOIDAL WAVES:

Digital signals are in form of rectangular waves as shown in figure on the next page. The rectangular wave can be decomposed in to superposition of sinusoidal waves of frequencies ν_0, $2\nu_0$, $3\nu_0$, $4\nu_0$ .... $n\nu_0$ where n tends to infinity, and ν_0 is fundamental frequency (i.e $\frac{1}{T_0}$ where T_0 is periodicity of rectangular wave).



Approximation of a rectangular wave in terms of a fundamental sine wave and its harmonics.

YOUTUBE.COM/@THEPHYSICSSHELTER

One can see from this figure that as we add more and more harmonics, we get closer and closer to the rectangular wave however to generate an exact rectangular wave we will need infinite number of harmonics and accordingly the signal bandwidth will be infinite. However for practical applications we can reasonably consider this as generated from a few harmonics and contribution from higher harmonics can be neglected, thus limiting the bandwidth. If the receiver approximates the rectangular pulses in this manner, we get a distorted version of a transmitted signal, however if the bandwidth is large enough, the information is not lost and the rectangular signal is recovered to a reasonable accuracy. (The reason is that higher the harmonic, lesser is its contribution to the waveform).

BANDWIDTH OF TRANSMISSION MEDIA: The range of frequencies which can be satisfactorily transmitted through a transmission medium defines its bandwidth. The commonly used transmission media are wires, coaxial cables fiber optic cables and free space. Their

bandwidths are described below.

COAXIAL CABLES: Coaxial cables have a bandwidth of about 750 MHz. These cables are normally operated at frequencies below 18 GHz.

OPTICAL FIBERS: Optical fibers can offer a transmission bandwidth of above 100 GHz and the typical operation frequencies in optical fiber communication range from 1 THz to 1000 THz, (i.e. microwaves to ultraviolet).

FREE SPACE: (WIRELESS COMMUNICATION)

SOME IMPORTANT WIRELESS COMMUNICATION FREQUENCY BANDS

Service	Frequency bands	Comments
Standard AM broadcast	540-1600 kHz	
FM broadcast	88-108 MHz	
Television	54-72 MHz	VHF (very high frequencies)
	76-88 MHz	TV
	174-216 MHz	UHF (ultra high frequencies)
	420-890 MHz	TV
Cellular Mobile Radio	896-901 MHz	Mobile to base station
	840-935 MHz	Base station to mobile
Satellite Communication	5.925-6.425 GHz	Uplink
	3.7-4.2 GHz	Downlink

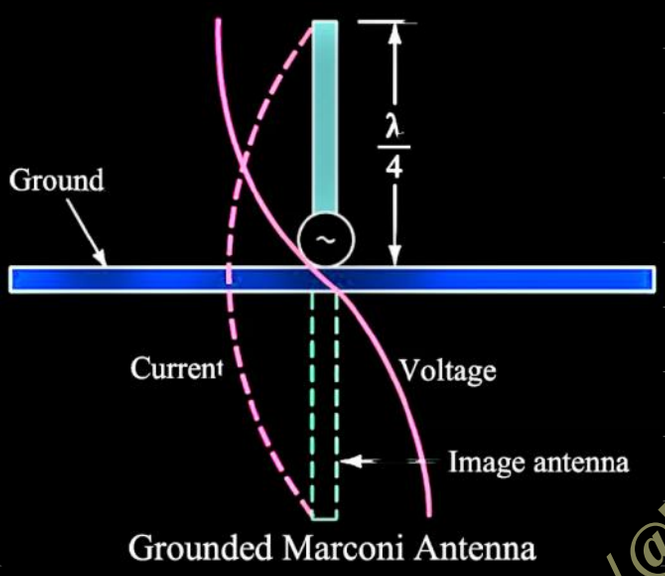
YOUTUBE.COM/@THEPHYSICSSHELTER

Communication through free space relies on radio waves ranging from few hundreds of kHz to few GHz. These frequencies are further sub divided and allocated for various services as indicated in the table above. The allocations of the spectrum for various types of communication are done according to an international agreement. The present system of frequencies is administered by the International Communication Union. A typical TV channel is allotted 6 MHz of bandwidth for transmission (though

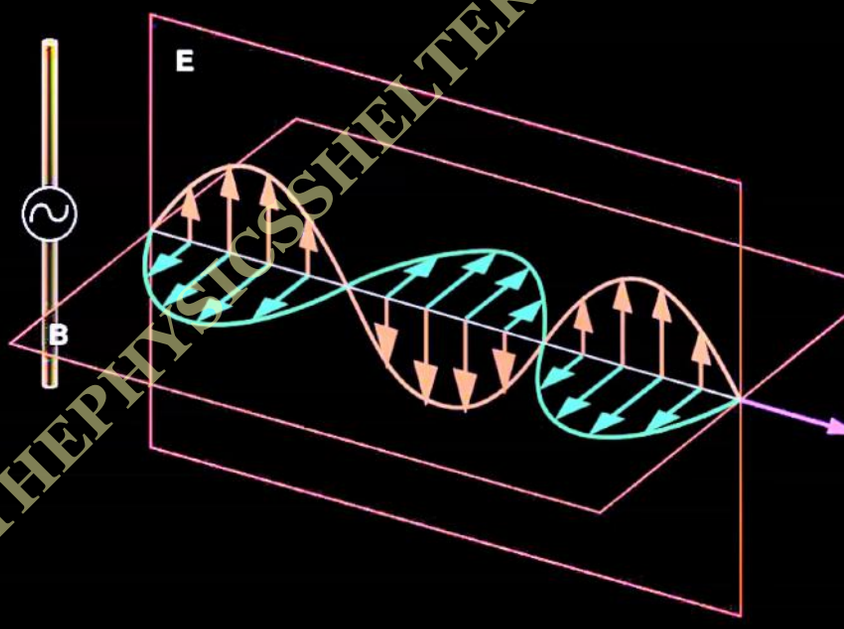
could vary between 6-8 MHz for various frequency bands). The number of TV channels that can be accomodated in any fuquency band can be calculated by dividing the width of a frequency band (say UHF) by the bandwidth required by each channel. Thus most TV channels will be accomodated in the last TV frequencies band i.e. 420-890 MHz (See table).

Propagation Of Electromagnetic Waves:

Antenna:



Grounded Marconi Antenna



An antenna emits electromagnetic waves being driven by a power source. Due to alternating current, the antenna behaves like a dipole of varying dipole moment, thus creating a time varying electric field in the surroundings. The minimum height of antenna required is $\lambda/4$ where λ is the wavelength of the EM waves. (i.e $\lambda = c/\nu$). Thus if transmission is done at low frequencies, antenna size required will be huge. For such transmissions, ground based vertical antennae are used. (The height of antenna is related to formation of stationary waves. Some antena have height $\lambda/2$ also).

Ground Wave The waves for amplitude modulate transmission (AM) are created by ground based vertical antennae. For such antennae, ground has a strong influence on propagation of signal. The waves glide over the surface of earth and this type of waves are called ground waves or surface waves.

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

The ground waves induce a current in the ground over which it passes and it is attenuated as a result of energy absorption by the earth. This attenuation of surface waves increases very rapidly with frequency. The maximum range of coverage of transmission is governed by frequency and transmitted powers. For ground wave based communication, typical frequency is less than a few MHz. For instance see the table of frequencies for wireless communication. It can be seen that standard AM broadcast uses frequencies from 0.54 MHz to 1.6 MHz.

Layers Of Atmosphere And Interactions With Em Waves:

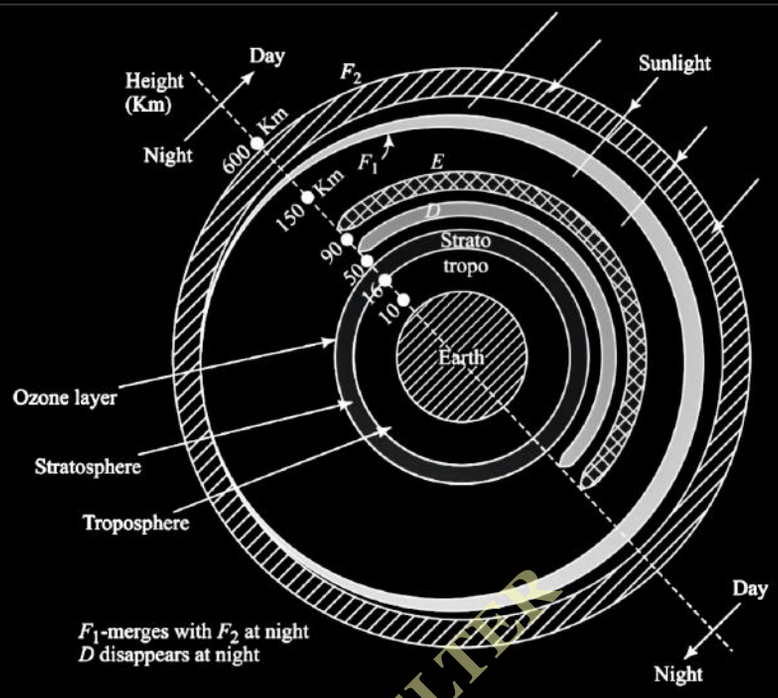
Different layers of atmosphere and their interaction with the propagating electromagnetic waves

Name of the stratum (layer)		Approximate height over earth's surface	Exists during	Frequencies most affected
Troposphere		10 km	Day and night	VHF (up to several GHz)
D (part of stratosphere)	PARTS OF IONOSPHERE	65-75 km	Day only	Reflects LF, absorbs MF and HF to some degree
E (part of Stratosphere)		100 km	Day only	Helps surface waves, reflects HF
F_1 (Part of Mesosphere)		170-190 km	Daytime, merges with F_2 at night	Partially absorbs HF waves yet allowing them to reach F_2
F_2 (Thermosphere)		300 km at night, 250-400 km during daytime	Day and night	Efficiently reflects HF waves, particularly at night

The electromagnetic waves used for communication pass through the atmosphere and depending on the frequency, the reflection and refraction behavior (also absorption) may vary for various frequency ranges and atmospheric composition. The ionosphere starts above 50 km and it consists of ions produced by α, β, γ rays as well as cosmic rays and meteors.

YOUTUBE.COM/@THEPHYSICSSHELTER

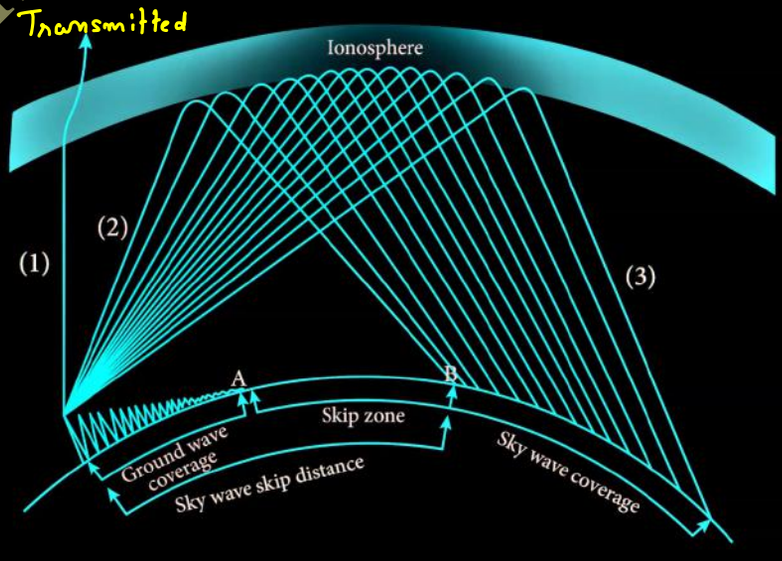
The four important layers of ionosphere are D, E, F_1 and F_2. The temperature of these layers changes over the day, over the year as well as sunspot cycle of 11 years etc. When EM waves enter various layers, it bends due to gradual reflection and then finally may undergo total internal reflection just like an optical ray.



The ion density of ionosphere keeps on increasing as we go deeper in to space. The refractive index and dielectric constant are functions of ion density (N)

Sky Waves:

The mode of propagation in which the electromagnetic waves radiated upward from antenna at large angles, gets reflected back towards earth is called sky wave propagation or ionospheric propagation. The corresponding waves are called sky waves. The frequency range for these waves is 3 to 30 MHz (NCERT says few MHz to 30 to 40 MHz). The waves of frequency more than 30 MHz can easily penetrate through the atmosphere and do not undergo any reflection. These waves are used for short wave broadcast services. Medium and high frequencies are used for long distance radio communication. Extremely long distance communication is also possible as the radio waves can undergo multiple reflections. A

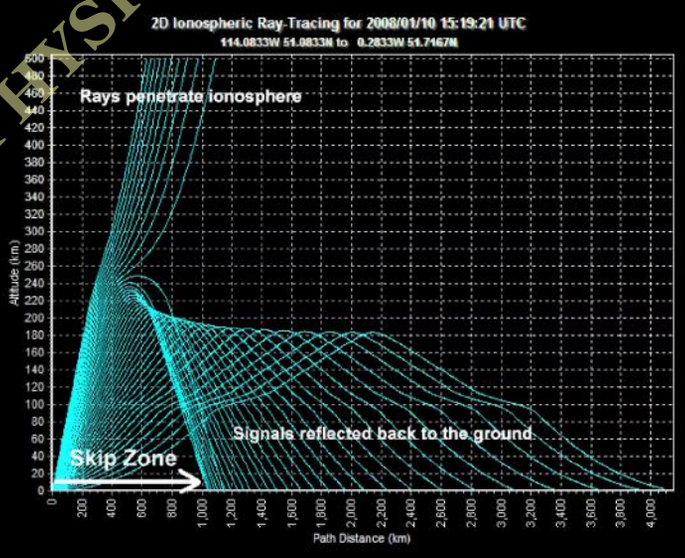


YOUTUBE.COM/@THEPHYSICSSHELTER

single reflection helps radio waves to travel a distance of about 4000 Km. Since sky wave communication is based on total internal reflection, the waves must be sent at angles greater than or equal to the critical angle to ensure that waves are reflected back to ground and do not escape to the space.

SKIP DISTANCE: The shortest distance between the transmitter and the point of reception of the sky wave along surface of earth is called skip distance (before this point, no sky waves can reach). (see the figure above).

SKIP ZONE: The region along surface of earth where neither sky waves nor ground waves can reach from a transmitter is called skip zone, or skip area. The skip zone is an annular area around the transmitter. Since reflection depends on frequency of the EM wave, skip zone can be reduced by decreasing frequency of EM waves used for communication. (By using suitable frequency it might also be possible to make the skip zone zero).



CRITICAL FREQUENCY:

It turns out that physics of propagation of EM waves in ionosphere is somewhat different from that of propagation in usual dielectric media. As EM wave propagates through space, it accelerates ions and electrons due to its electromagnetic field. This causes conduction as well as displacement currents in the ionosphere. The effective refractive index for ionosphere comes out to be $\mu = (1 - 81N/f^2)^{1/2}$ where N is density of electrons in the atmosphere and f is the frequency. Note that $\mu < 1$ but still the waves do not travel at speed greater than light.

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

This is just some mathematical equivalent μ for some specific kind of calculation purposes (beyond scope of JEE). For ionospheric reflection, refractive index must be complex, which places an upper limit on frequency. Thus, upper limit is given as:

$$f_c = 9\sqrt{N} \quad (1)$$ ← Critical frequency (upper limit).

N: Electron Density in ionosphere.

MAXIMUM USABLE FREQUENCY:

The highest frequency of radio waves which when sent at some particular angle of incidence towards ionosphere get reflected and return towards earth is called maximum usable frequency.
If f_c is the critical frequency, and θ is the angle of incidence then maximum usable frequency is given by:

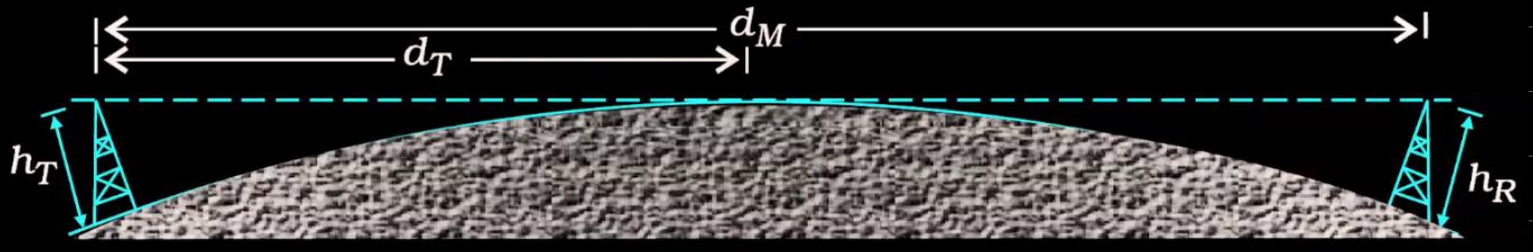
$$MUF = f_c / \cos\theta$$

→ This can be derived by putting $r = 90°$ and $i = \theta$ in snell's law using $\mu = (1 - 81N/f^2)^{1/2} = \left(1 - \left(\frac{f_c}{f}\right)^2\right)^{1/2}$.

SPACE WAVE: The electromagnetic waves of frequencies above 30 MHz are called space waves (NCERT: 40 MHz), and communication achieved through these waves is called space wave communication. At such high frequencies, the waves travel in straight line therefore such communication is also called line of sight communication.
For high frequencies the transmission towers must be high enough so that the transmitted and recieved signals do not encounter the curvature of earth (Waves travelling near surface face greater attenuation). The communication systems like television telecast, satellite communication, RADAR and microwave links are examples of communication systems that use space wave mode of propagation.

YOUTUBE.COM/@THEPHYSICSSHELTER

Horizon Of An Antenna And Maximum Distance Between Antennas :



Line of sight communication by space waves.

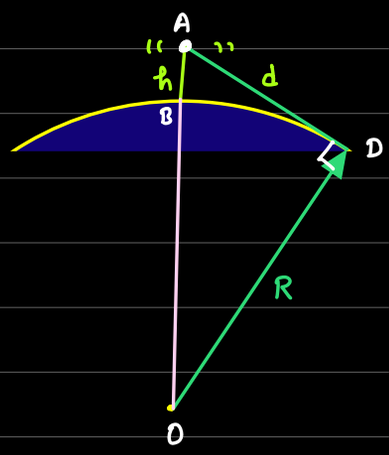
If the transmitting antenna has a height h_T, it can be shown that distance to its horizon (d_T) is given as $\boxed{d_T = \sqrt{2Rh_T}}$ where h_T is height of transmitting antenna and R is the radius of earth. Similarly the horizon distance of receiving antenna is $d_R = \sqrt{2Rh_R}$. Accordingly maximum possible line of sight distance between two antennas (d_M) having heights h_R and h_T above earth is given by

$$\boxed{d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}}$$

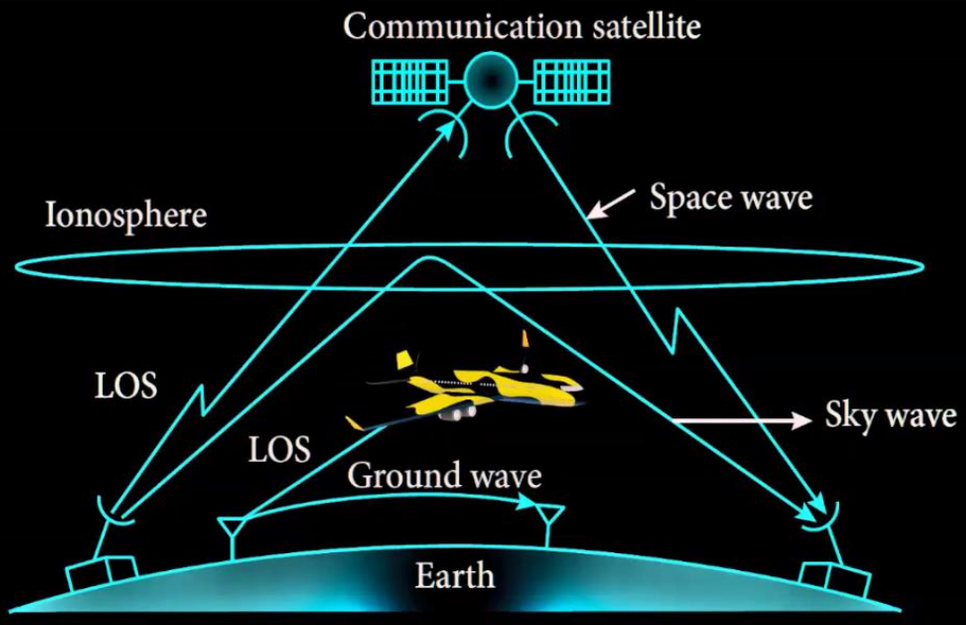
Proof : Consider the adjacent figure. Here, $h << R$ where h is height of the antenna. Now by Pythagoras theorem, $AD^2 = OA^2 - OD^2$

$\Rightarrow d^2 = (R+h)^2 - R^2 = 2Rh + h^2$ by $h << R$

thus, $d = \sqrt{2Rh}$ QED.



→ Summary of various modes of propagation of waves used for communication is presented in adjacent figure.



YOUTUBE.COM/@THEPHYSICSSHELTER

Example: NCERT: 15.1

Example 15.1 A transmitting antenna at the top of a tower has a height 32 m and the height of the receiving antenna is 50 m. What is the maximum distance between them for satisfactory communication in LOS mode? Given radius of earth 6.4×10^6 m.

Solution

$$d_m = \sqrt{2 \times 64 \times 10^5 \times 32} + \sqrt{2 \times 64 \times 10^5 \times 50} \text{ m}$$

$$= 64 \times 10^2 \times \sqrt{10} + 8 \times 10^3 \times \sqrt{10} \text{ m}$$

$$= 144 \times 10^2 \times \sqrt{10} \text{ m} = 45.5 \text{ km}$$

Base Band Signal: A message signal before being processed for transmission is called base band signal. Base band refers to the range of frequencies that constitute a message signal. In general the signals are not made of single sinusoidal frequency, rather they are superposition of several sinusoids as discussed earlier. The range of frequencies contained in the signal is called bandwidth. For instance for audio frequency signals, the frequencies range from 20 Hz to 20 KHz so we can say baseband width is ≈ 20 KHz

Modulation: Encoding of low frequency baseband signal in to a high frequency signal by suitably combining it with a high frequency carrier wave to make it suitable for transmission is called modulation.

Necessity Of Modulation:

There are three main reasons as to why modulation of baseband signals is required:

1. **Size Of Antenna:** As discussed earlier, the minimum size of antenna required for transmission of a wave is $\lambda/4$ where λ is the wavelength. Thus if transmission is done at a low frequency we will have to use very huge antennas. For instance, for a wave of frequency 20 KHz, the wavelength is 15 Km,

YOUTUBE.COM/@THEPHYSICSSHELTER

which would require an antenna of minimum height 3·75 Km. This is impractical. If instead, the frequency were 1 MHz, the required antenna height will reduce to only 75 m).

2. Effective Power Radiated By Antenna: A theoretical study of radiation from a linear antenna of length l shows that power radiated by an antenna is proportional to $(l/\lambda)^2$. Thus for antenna of same length, radiated power increased by decreasing the wavelength. For a good transmission over long distances, we need high power. Therefore we use high frequencies for modulating baseband signals.

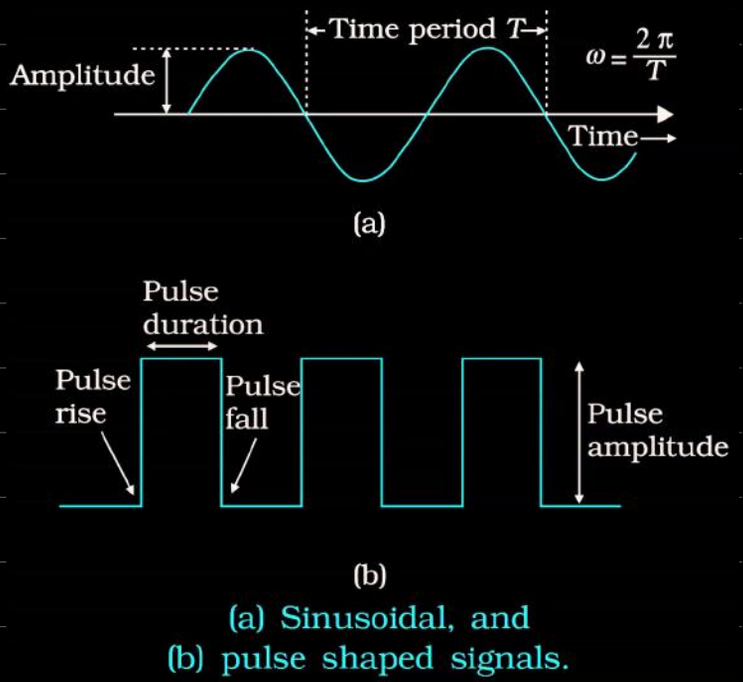
3. Mixing Up Of Baseband Signals: Suppose various transmitters were all transmitting simultaneously in the baseband frequencies, then their signals will get mixed up and there will be no simple way of distinguishing them, however if they were encoded using different carrier frequencies, we could tune the receiving antenna to receive the encoded signal from a particular radio station.

Two Types Of Carrier Waves:

As described in previous section, carrier waves are high frequency waves used for encoding message signals making them suitable for transmission. There are two basic types of carrier waves which are used for this purpose.

Sinusoidal Waves:

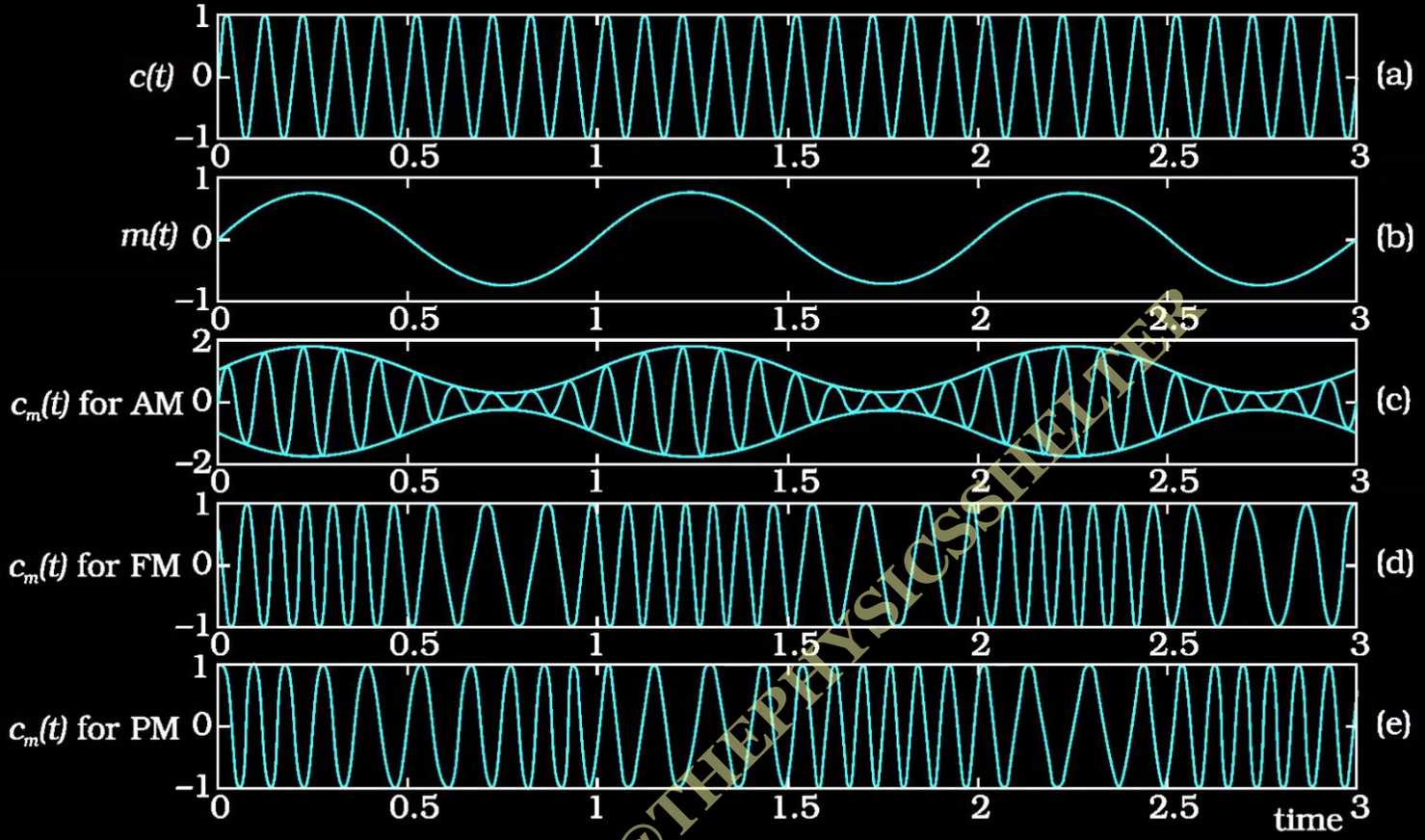
A sinusoidal carrier wave can be represented as $C(t) = A_c \sin(\omega_c t + \phi)$ where $C(t)$ could represent a voltage or current in the equipment generating the carrier waves. Here A_c is the amplitude of the carrier, $\omega_c (= 2\pi\nu_c)$ is the angular frequency and ϕ is the initial phase. During the process of modulation, any of the three parameters namely A_c, ω_c or ϕ can be controlled by the message or information



(a) Sinusoidal, and
(b) pulse shaped signals.

YOUTUBE.COM/@THEPHYSICSSHELTER

signal. (correspondingly we call the modulation type being Amplitude modulation (AM), frequency modulation (FM) or phase modulation (PM). The various types of modulations for a single tone (single frequency) message signal are illustrated in the following figure.



Modulation of a carrier wave: (a) a sinusoidal carrier wave; (b) a modulating signal; (c) amplitude modulation; (d) frequency modulation; and (e) phase modulation.

YOUTUBE.COM/@THEPHYSICSSHELTER

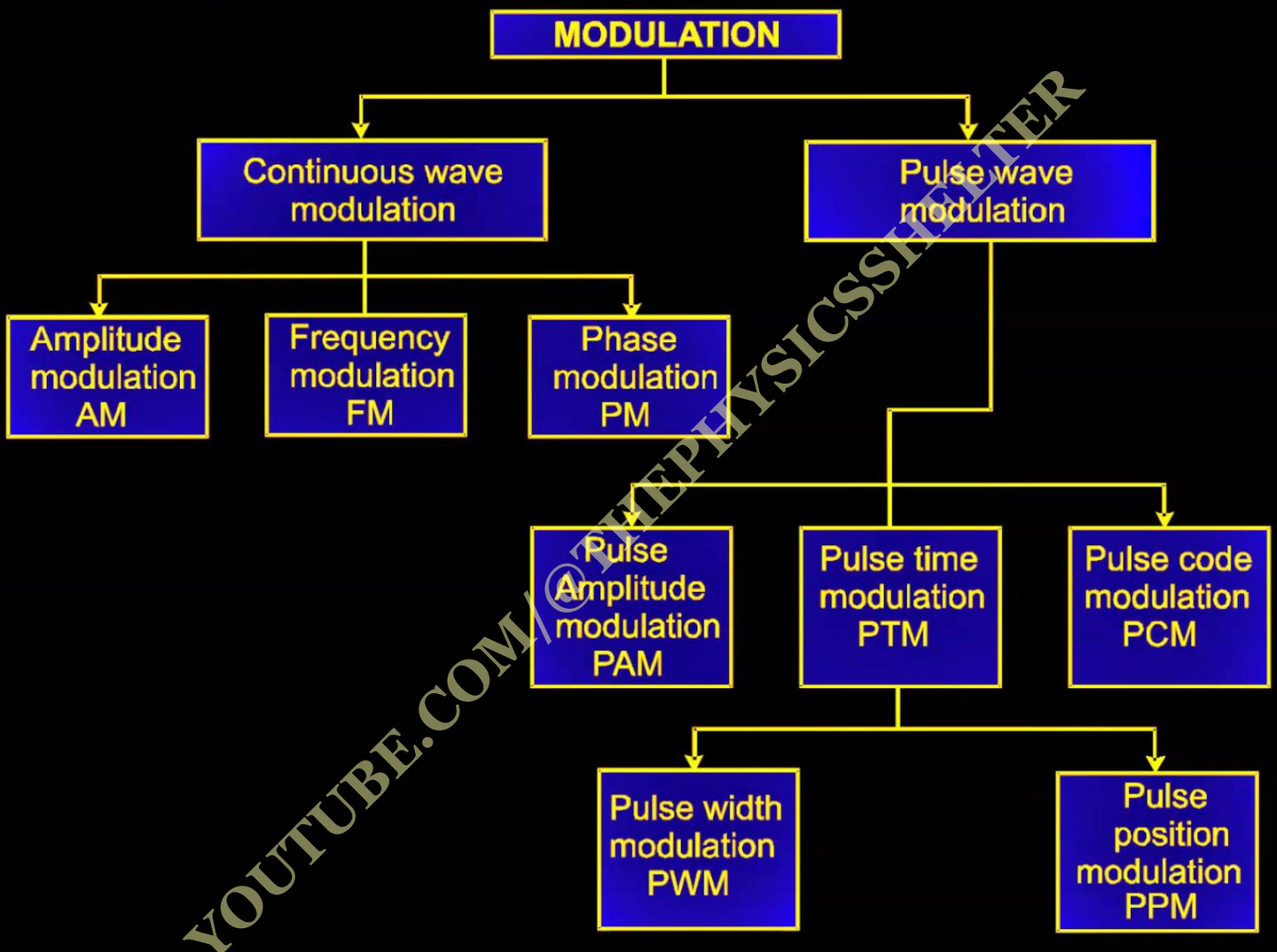
Notice that when message signal increases, in frequency modulation, the frequency increases while in amplitude modulation, the amplitude of transmitted wave increases. The functional forms of phase modulated and frequency modulated waves are similar though algorithm for obtaining phase modulated wave is different. (Just by looking at functional form of a wave we cannot say whether it is a frequency modulated wave or a phase modulated wave).

PULSE SHAPED CARRIER WAVES:

In case of pulse shaped carrier waves, the significant characteristics are: pulse amplitude, pulse duration (or pulse width) and pulse position (denoting the time when pulse begins to rise or fall).

Accordingly we have the corresponding modulation types namely (a) Pulse amplitude modulation (PAM) (b) Pulse duration modulation (PDM) or pulse width modulation (PWM) and (c) pulse position modulation (PPM). Besides these encodings there is also one more type of encoding known as Pulse Code Modulation (PCM).

→ All various types of modulations are summarized in the chart below:



YOUTUBE.COM/@THEPHYSICSSHELTER

AMPLITUDE MODULATION: Let the carrier wave be given by $C(t) = A_c \sin(\omega_c t)$ and the modulating signal (message signal) be represented by $m(t) = A_m \sin(\omega_m t)$ where $\omega_m = 2\pi f_m$ is angular frequency of the message signal (assumed single tone). Then in case of amplitude modulation, the desired wave for transmission is given as:

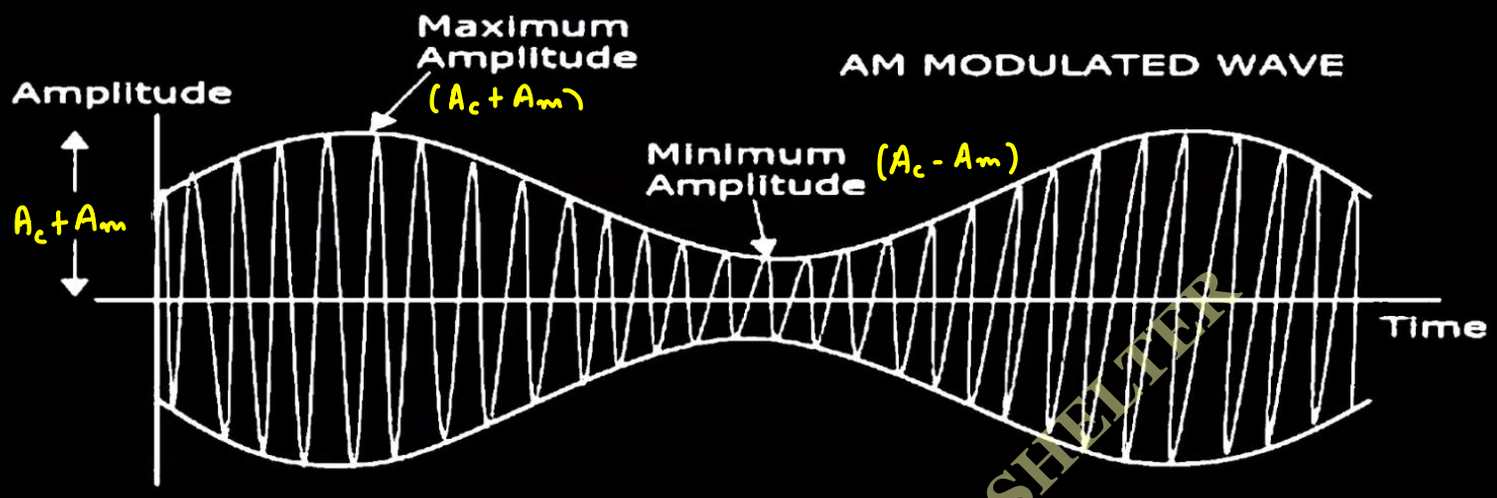
$$C_m(t) = [A_c + A_m \sin(\omega_m t)] \sin \omega_c t \quad (1)$$

AMBARISH SIR

https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

MODULATION INDEX: The ratio of amplitudes of modulating signal to that of carrier signal is called modulation index (μ).

Mathematically, $\boxed{\mu = \frac{A_m}{A_c}}$



From equation ① above it can be seen that maximum and minimum amplitudes of modulated wave are given as:

$A_{max} = A_c + A_m$ and $A_{min} = A_c - A_m$ thus we can also express the modulation index as:

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

→ For achieving a good transmission modulation index is always kept less than unity (i.e $\mu < 1$). If $\mu \geq 1$ then the condition is called overmodulation and it leads to distortion and a noisy signal ($\mu \geq 1$ would mean that at times amplitude would go to zero which means power of the wave would go to zero. At this instant all we recieve is the noise).

SIDE BANDS AND BANDWIDTH: Equation ① can be rewritten as:

$$C_m(t) = A_c\left(1 + \frac{A_m}{A_c}\sin\omega_m t\right)\sin\omega_c t = A_c\left(1 + \mu\sin\omega_m t\right)\sin\omega_c t$$

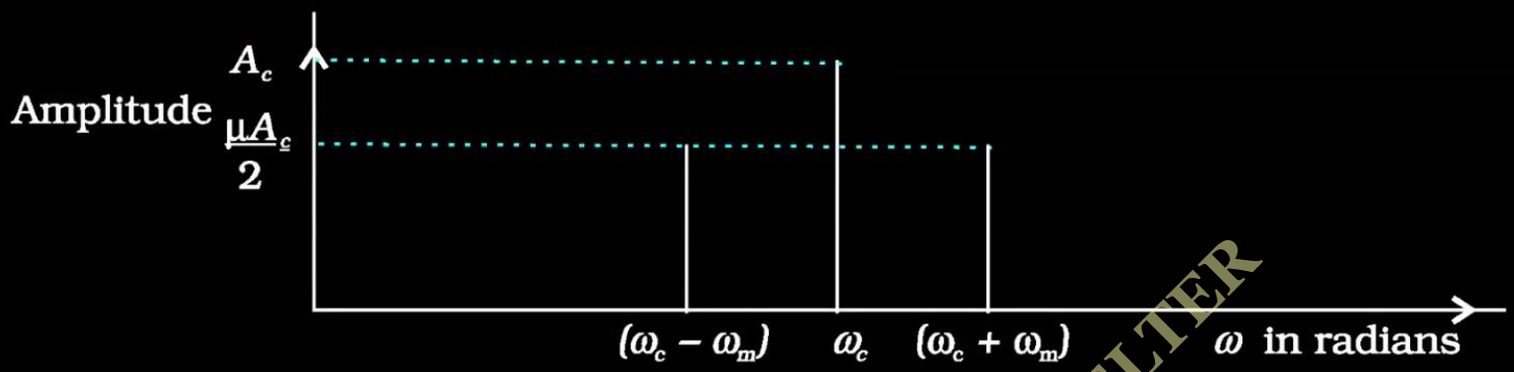
$$= A_c \sin\omega_c t + \mu A_c \sin\omega_m t \sin\omega_c t.$$

Now using $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$,

YOUTUBE.COM/@THEPHYSICSSHELTER

$$c_m(t) = A_c \sin \omega_c t + \frac{\mu A_c}{2} \cos(\omega_c - \omega_m)t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_m)t.$$

Thus we can see the net transmitted wave as superposition of three sinusoids having frequencies ω_c, $\omega_c - \omega_m$ and $\omega_c + \omega_m$.



A plot of amplitude versus ω for an amplitude modulated signal.

A plot of amplitudes vs frequencies in a signal is called frequency spectrum plot. The above figure shows the frequency spectrum plot of Amplitude modulated single tone wave.

→ $\omega_{min} = \omega_c - \omega_m$ is called lower side band frequency (LSB)

→ $\omega_{max} = \omega_c + \omega_m$ is called upper side band frequency (USB).

A particular radio station will operate at a certain ω_c and occupy a bandwidth $\omega_{max} - \omega_{min}$ i.e $2\omega_m$ around the central carrier frequency. While allotting carrier frequencies to various radio stations, care is taken that their side bands should not overlap. As long as the broadcast frequencies (carrier waves) are sufficiently spaced out so that their side bands do not overlap, the radio stations can operate without interfering with each other).

NCERT EXAMPLE: 15.2:

Example 15.2 A message signal of frequency 10 kHz and peak voltage of 10 volts is used to modulate a carrier of frequency 1 MHz and peak voltage of 20 volts. Determine (a) modulation index, (b) the side bands produced.

Solution

(a) Modulation index =10/20 = 0.5

(b) The side bands are at (1000+10 kHz)=1010 kHz and (1000 –10 kHz) = 990 kHz.

YOUTUBE.COM/@THEPHYSICSSHELTER

Amplitude Modulation With Multitone Message Signal :

If a carrier wave is modulated using a superposition of multiple sine waves with individual modulation indices being m_1, m_2 and m_3 etc then the total modulation index is given as:

$$m = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$$

its derivation is based on a similar definition of modulation index i.e

m = Maximum Amplitude of Modulating Signal / (Amplitude of Carrier).

We are omitting the derivation here (it is part of any standard course on signal processing or communication systems at UG level).

Here: $m_1 = A_{m1}/A_c$, $m_2 = A_{m2}/A_c$ etc...

Power In AM Waves: In any alternating current circuit if there are several sources having distinct frequencies from each other then average power supplied to the circuit under their simultaneous action is simply summation of their individual powers considered one at a time. (This has to do with the fact that $\sin\omega_1 t \times \sin\omega_2 t$ has an average value of zero when $\omega_1 \neq \omega_2$). Now any amplitude modulated wave can be expressed as superposition of several sinusoidal waves (e.g. we saw for a single tone modulating function). Thus net power can also be written here as summation of powers due to each individual term.

Carrier Power: Recall that in an AC circuit power is V_{rms}^2/R thus for carrier wave acting alone, carrier power is given as

$$P_c = \frac{(A_c/\sqrt{2})^2}{R} = \frac{A_c^2}{2R}, \text{ thus } \boxed{P_c = \frac{A_c^2}{2R}}$$

Lower Side Band Power: The amplitude of lower side band is $\mu A_c/2$ thus Power associated with lower side band is:

$$P_L = (\mu A_c/2\sqrt{2})^2/2R = \mu^2 A_c^2/8R$$

Upper Side Band Power For upper side band also the

YOUTUBE.COM/@THEPHYSICSSHELTER

amplitude is $\mu A_c/2$ thus upper side band power is $P_U = \mu^2 A_c^2/8R$.

TOTAL SIDE BAND POWER: The sum of powers of lower side band and upper side band is called total side band power. Thus:

$$P_{SB} = \mu^2 A_c^2/4R = (\mu^2/2) P_c$$

TOTAL TRANSMITTER POWER: The sum of carrier power and side band powers is the net power of the transmitter thus net transmitter power:

$$P_T = P_c + P_{SB} = \frac{A_c^2}{2R}\left[1 + \frac{\mu^2}{2}\right] \quad \text{thus:}$$

$$P_T = \frac{A_c^2}{2R}\left[1 + \frac{\mu^2}{2}\right] = P_c\left[1 + \frac{\mu^2}{2}\right]$$

TRANSMISSION EFFICIENCY: When we expressed the modulated signal as sum of three sinusoidal terms, one of the terms was the carrier wave itself which did not contain ω_m and the other two terms were side band terms which contained ω_m. Thus our useful information is contained only in side bands. The ratio of side band power to the total transmitter power is called transmission efficiency (η). Thus:

$$\eta = \frac{P_{SB}}{P_T} = \frac{(\mu^2/2) P_c}{(1+\mu^2/2) P_c} = \frac{\mu^2}{2+\mu^2}$$

Thus: $\eta = \mu^2/(2+\mu^2)$

→ Maximum transmission efficiency without signal distortion occurs for $\mu = 1$ thus $\eta_{max} = 33\%$.

→ Modulation index is alternatively also known as modulation factor, degree of modulation or modulation coefficient. Sometimes it is also expressed in percentage (called percent modulation).

YOUTUBE.COM/@THEPHYSICSSHELTER

Transmitter Current In Terms Of Carrier Current: When only carrier is present we call the current as carrier current while the current corresponding to the modulated wave is called transmitter current. Since power is proportional to square of current we can write the relation between transmitter current (I_t) and carrier current (I_c) as:

$$\frac{P_t}{P_c} = \frac{I_t^2}{I_c^2} = \left(1 + \frac{\mu^2}{2}\right) \Rightarrow \boxed{\frac{I_t}{I_c} = \sqrt{1 + \frac{\mu^2}{2}}}$$

Example: Antenna current of an AM transmitter is 8A when only carrier is present and it increases to 8.93 A if the carrier is modulated by a single tone sinusoidal message signal. (a) Find the percent modulation (b) Also find the antenna current if the percent modulation changes to 80 %.

Solution: $I_c = 8A$, $I_t = 8.93A \Rightarrow \frac{8.93}{8} = \sqrt{1 + \mu^2/2}$

Solving for μ, we get, $\mu \approx 0.701 \Rightarrow 70.1\%$ modulation. Ans (a).

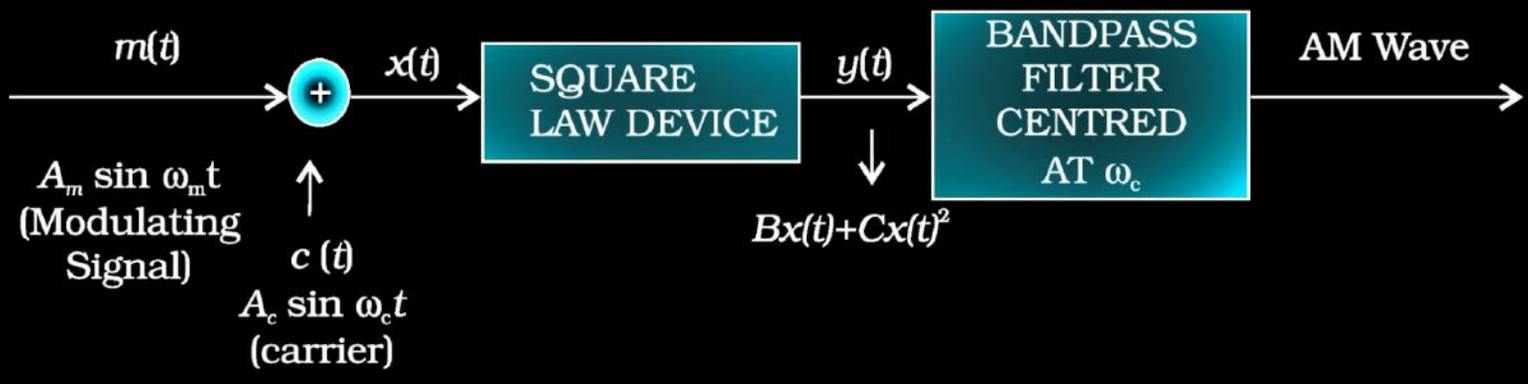
Now with 80% modulation, $\mu = 0.8 \Rightarrow I_t = I_c\sqrt{1 + \frac{\mu^2}{2}} = 8\sqrt{1 + \frac{0.64}{2}}$

$\Rightarrow I_t = 9.2\ A$ Ans (b)

Limitations Of Amplitude Modulation: Some of the disadvantages of amplitude modulation are, (a) Noisy Reception (b) low efficiency (c) Small operating range and (d) Poor audio quality. Infact because of these limitations, the use of amplitude modulation has severely reduced. Now a days, most of the radio stations operate on frequency modulation technology.

YOUTUBE.COM/@THEPHYSICSSHELTER

Production Of Amplitude Modulation Wave:



Block diagram of a simple modulator for obtaining an AM signal.

In this section we will not be learning about the circuitry involved in producing Amplitude modulated wave, rather here we will be learning about various mathematical operations achieved through the circuitry and how it leads to production of Amplitude Modulated waves. Here modulating signal $A_m \sin(\omega_m t)$ is added to the carrier signal $A_c \sin \omega_c t$ to produce the signal $x(t)$. This signal $x(t) = A_m \sin(\omega_m t) + A_c \sin(\omega_c t)$ is passed through a square law device which is a non linear device producing an output of the form $y(t) = B\,x(t) + C\,x^2(t)$ where B and C are some constants. Thus,

$$y(t) = BA_m \sin \omega_m t + BA_c \sin \omega_c t$$
$$+C\left[A_m^2 \sin^2 \omega_m t + A_c^2 \sin^2 \omega_c t + 2A_m A_c \sin \omega_m t \sin \omega_c t\right]$$
$$= BA_m \sin \omega_m t + BA_c \sin \omega_c t$$
$$+\frac{CA_m^2}{2} + \frac{CA_c^2}{2} - \frac{CA_m^2}{2}\cos 2\omega_m t - \frac{CA_c^2}{2}\cos 2\omega_c t$$
$$+ CA_m A_c \cos(\omega_c - \omega_m)\,t - CA_m A_c \cos(\omega_c + \omega_m)\,t$$

where the trigonometric relations $\sin^2 A = (1 - \cos 2A)/2$ and the relation for $\sin A \sin B$ mentioned earlier are used.

Here there is a dc term $\frac{C}{2}(A_m^2 + A_c^2)$, the low frequency term involving frequencies ω_m and $2\omega_m$, a very high frequency terms of frequency $2\omega_c$ and the terms required for AM wave with frequencies ω_c, $\omega_c - \omega_m$ and $\omega_c + \omega_m$.

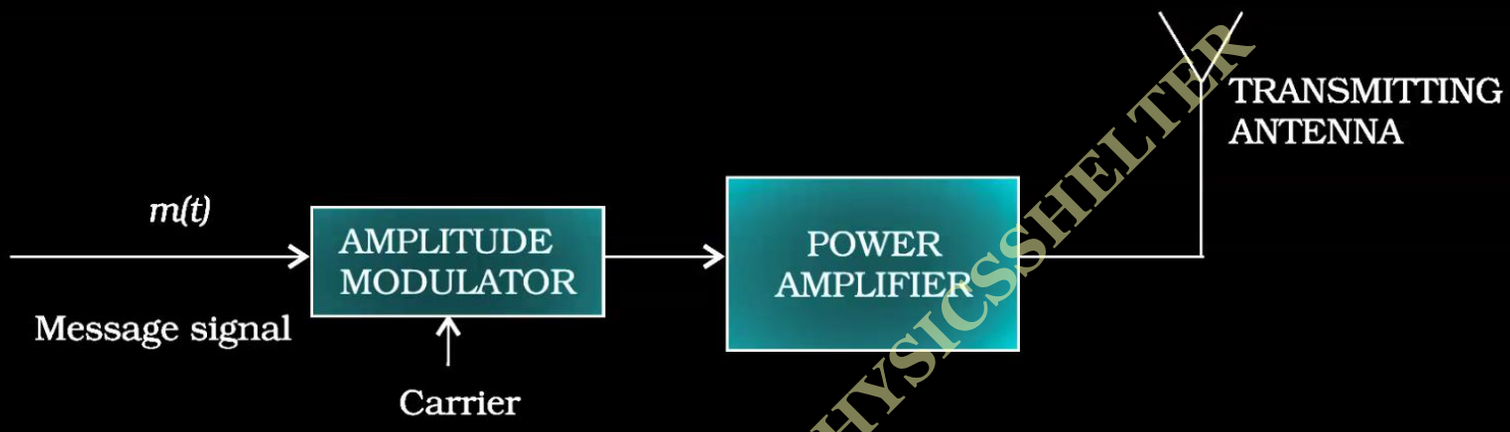
YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Now we pass this signal through a band pass filter centred at ω_c which cuts out terms with very high and very low frequencies and we are left with the output:

$$BA_c \sin\omega_c t + CA_mA_c \cos(\omega_c - \omega_m)t - CA_mA_c \cos(\omega_c + \omega_m)t$$

This is exactly of the same form as the required amplitude modulated wave.

Block Diagram Of A Transmitter:



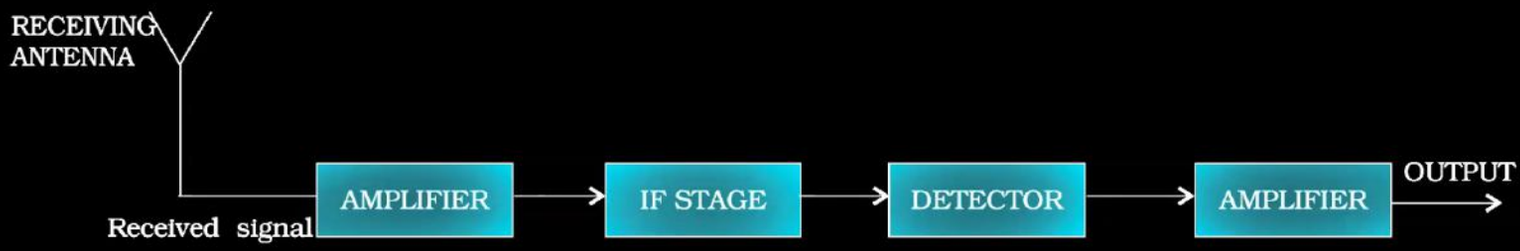
Block diagram of a transmitter.

The modulated signal cannot be transmitted directly because to send the signal through large distances the signal should have sufficient power. Therefore the output of the amplitude modulator is sent to a power amplifier (such as the one's we studied in chapter on semi-conductors) and then the output of Power amplifier is sent to the transmitting antenna, which radiates the signal in to the surrounding space in the form of EM waves. The modulator, the Power amplifier and the Transmitting Antenna together form the transmitter.

Detection Of Amplitude Modulated Waves:

The block diagram of a receiver is shown on the next page. The transmitted waves attenuate in propagating through the channel. Therefore, the receiving antenna is followed by an amplifier, and a detector. In addition, to facilitate an efficient detection of the signal, the high frequency signal is converted to an intermediate frequency using a set up called Intermediate frequency stage (IF STAGE).

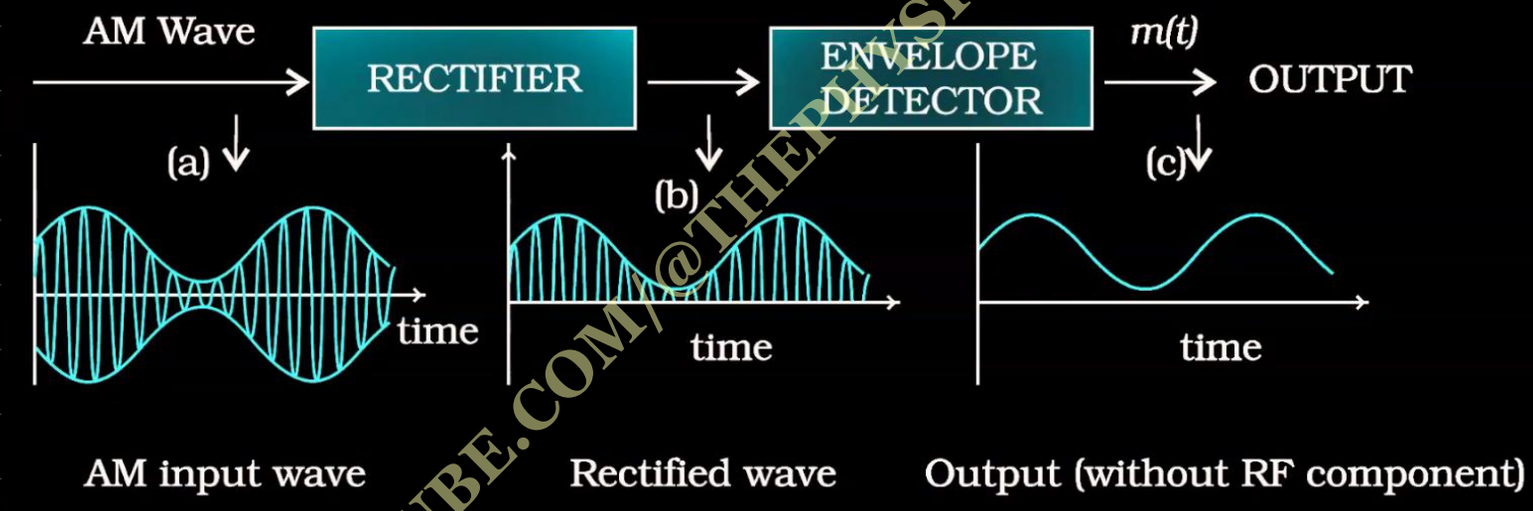
YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter



Block diagram of a receiver.

The IF stage is followed by a detector which recovers the message signal from the modulated signal. Recall that message signal forms the envelope of the modulated signal so the device that we use for retrieving the message is called envelope detector.

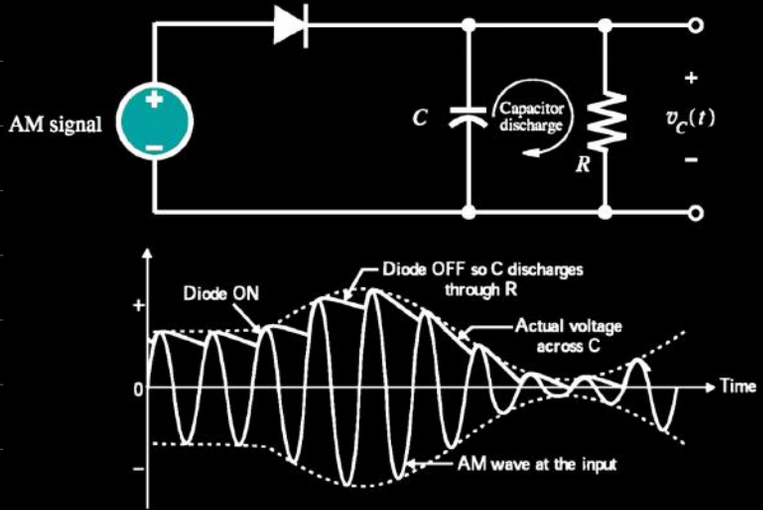
ENVELOPE DETECTOR:



Block diagram of a detector for AM signal. The quantity on y-axis can be current or voltage.

YOUTUBE.COM/@THEPHYSICSSHELTER

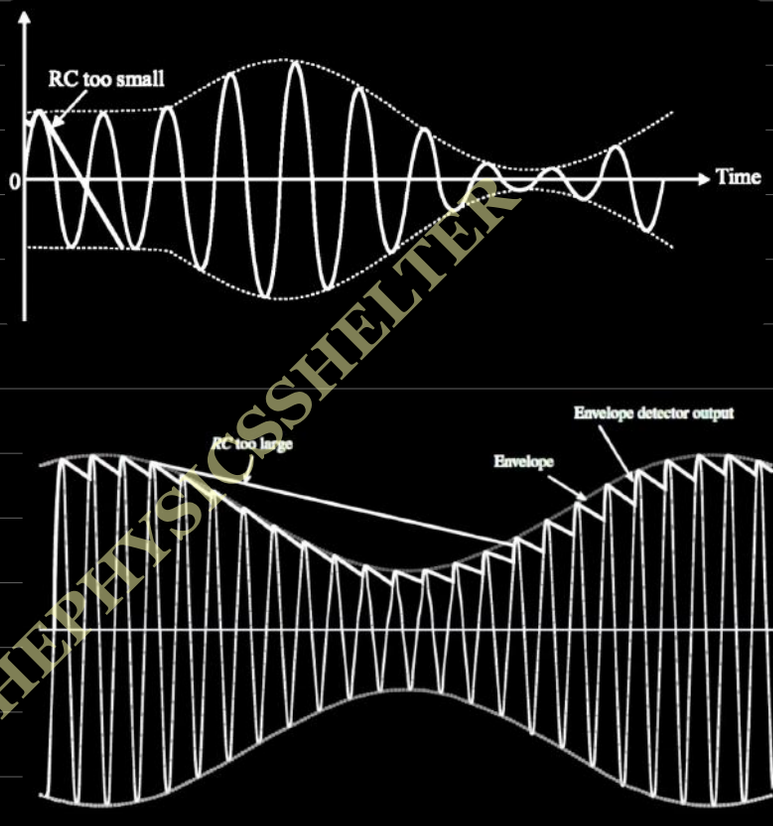
An envelope detector typically relies on a rectifier like diode circuit followed by an RC filter. When the modulated signal reaches near its peak, the diode get switched on and output equals the peak voltage. When the output voltage dips below the envelope, the capacitor begins to



discharge through an RC circuit of an optimal time constant until once again the signal reaches near the peak and diode gets switched on again (when signal voltage exceeds the capacitor voltage). Thus we can extract the waveform near the peaks of the AM signal.

Optimal Time Constant Of RC Circuit :

For getting a good detection of the envelope we need to choose an optimal value of the time constant RC of the filter circuit. If the time constant is too small, the diode won't turn off even when the waveform has significantly dipped from the envelope, whereas if RC is too large the diode won't turn on when the signal is near the envelope at a peak smaller than the previous peak. For a good detection it can be shown that :



$$\frac{1}{\omega_c} \leq RC \leq \frac{\sqrt{1-\mu^2}}{\mu \omega_m}$$

As a rule of thumb, generally RC is chosen such that :

$$\frac{1}{\omega_c} << RC << \frac{1}{\omega_m}$$

Example : JEE Main 2013 :

A diode detector is used to detect an amplitude modulated wave of 60% modulation by using a capacitor of 250 pF in parallel with load resistance of 100 KΩ. Find the maximum modulating frequency that could be detected by it. Also find the minimum carrier frequency.

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution:

$$\frac{1}{\omega_c} \leq RC \leq \frac{\sqrt{1-\mu^2}}{\mu \omega_m}$$

$$\Rightarrow \quad \omega_m \leq \frac{\sqrt{1-\mu^2}}{\mu RC} \quad \Rightarrow \quad f_m \leq \frac{\sqrt{1-\mu^2}}{2\pi \mu RC}$$

Now substituting the values: $f_m \leq 8488.26\ Hz = 8.48\ kHz$ Ans.
(Best option in JEE main was 5.31 kHz).
Now for minimum carrier frequency, $\omega_c \geq \frac{1}{RC} \Rightarrow f_c \geq \frac{1}{2\pi RC}$

$\Rightarrow f_c \geq 6366.2\ Hz = 6.4\ KHz$. Thus minimum carrier frequency must be 6.4 KHz.

Frequency Modulation:

In amplitude modulation we saw that amplitude of modulated signal varied with time thus leading to a fluctuating power associated with the transmitted signal and hence causing noisy reception.
FM takes care of these problems by maintaining a constant Amplitude and having only small fractional changes in the frequency of transmitted signals.

Meaning Of Instantaneous Frequency:

Suppose a signal is given as $A \sin[\theta(t)]$ then its instantaneous frequency is defined as:

$$\boxed{f(t) = \frac{1}{2\pi}\frac{d\theta}{dt}}$$

one can readily see that this definition also fits the functions with constant frequencies (e.g. $C(t) = A_c \sin(\omega_c t)$ will have $f_c = \omega_c/2\pi$).

Instantaneous Frequency In Frequency Modulation:

Let modulating signal be given as $m(t) = A_m \cos(2\pi f_m t)$
and let carrier signal be given as $C(t) = A_c \cos(2\pi f_c t)$.

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

The instantaneous frequency of the FM wave is adjusted such that

$\frac{1}{2\pi}\frac{d\theta}{dt} = f_i(t) = f_c + K_f m(t)$ where K_f is called sensitivity.

Thus $f_i(t) = f_c + K_f A_m \cos(2\pi f_m t) = \frac{1}{2\pi}\frac{d\theta}{dt}$ ①

→ Be careful, here we are assuming K_f in units of Hertz / Volt, if K_f were given in radians/voltsecit would mean $\omega_i(t) = \omega_c + K_f A_m \cos(\omega_m t)$.

FREQUENCY DEVIATION : The maximum change in frequency from the mean value is called frequency deviation. In this case, mean frequency is f_c and minimum and maximum frequencies are $f_c + K_f A_m$ and $f_c - K_f A_m$ that frequency deviation is given as

$$\boxed{\delta = K_f A_m}$$ (or $K_f A_m/2\pi$ in case K_f is in rad/volt etc.

FUNCTIONAL FORM OF FREQUENCY MODULATED WAVE :

If we consider the frequency modulated wave as $C_m(t) = A_c \cos(\theta)$
Then θ can be found by integrating equation ①.
Accordingly,

$$\frac{1}{2\pi}\frac{d\theta}{dt} = f_c + K_f m(t)$$

$$\Rightarrow \theta = 2\pi f_c t + 2\pi K_f \int m(t)\,dt$$

Accordingly the modulated wave becomes

$$\boxed{C_m(t) = A_c \cos\left(2\pi f_c t + 2\pi K_f \int m(t)\,dt\right)}$$

Now putting the value of $m(t)$, our function becomes

$$C_m(t) = A_c \cos\left(2\pi f_c t + \frac{2\pi K_f A_m}{2\pi f_m}\sin(2\pi f_m t)\right)$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Thus $C_m(t) = A_c \cos\left(2\pi f_c t + \frac{K_f A_m}{f_m} \sin(2\pi f_m t)\right)$.

MODULATION INDEX:

The ratio of frequency deviation to the frequency of modulating signal is called modulation index. Thus:

$$\mu = \frac{\delta}{f_m} = \frac{K_f A_m}{f_m}$$

CARRIER SWING: The difference in maximum and minimum frequency in the final modulated wave is called carrier swing. Now $f_{max} = f_c + \delta$ and $f_{min} = f_c - \delta$ thus carrier swing:

$$CS = 2\delta$$

FREQUENCY SPECTRUM: FM side band modulated signal consists of infinite number of side bands whose frequencies are $(f_c \pm f_m)$, $(f_c \pm 2f_m)$, $(f_c \pm 3f_m)$... However not all bands are significant. Around 98% of energy is concentrated in only a few bands around the central carrier frequency. These are known as significant side bands. (Note that this spectrum arises when we try to write FM signal as sum of simple sinusoids).

CARSON'S RULE: Number of significant side band pairs around a central frequency is $\mu + 1$, thus total number of significant side bands is $2(\mu+1)$, and the bandwidth then becomes $2(\mu+1)f_m$.

→ Number of significant side bands: $n = 2(\mu+1)$

→ Bandwidth assuming only significant side bands $B.W. = 2(\mu+1)f_m$

EXAMPLE: In an FM set up carrier frequency is 500 KHz, the modulating frequency is 15 kHz and frequency deviation is 75 kHz. Find (a) Modulation index (b) Number of side bands (c) Band Width.

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: $\Delta f = 75\ kHz$, $f_m = 15\ KHz \Rightarrow \mu = \Delta f / f_m = 5$ Ans.

Now number of side bands, $n = 2(\mu+1) = 12$ Ans.

Band Width, $BW = 2(\mu+1) f_m = 12 \times 15 = 180\ KHz$ Ans.

DEVIATION RATIO: As discussed earlier, an FM wave requires infinite number of side bands, however in practice we limit to a certain highest/lowest pair of frequencies (say f_H & f_L). The ratio of deviation from carrier frequency for these frequencies to the modulating frequency is called deviation ratio. Thus

Deviation Ratio: $$D.R = \left(\frac{f_H - f_c}{f_m}\right)$$

EXAMPLE: JEE MAIN 2022: An FM broadcast transmitter uses a modulating frequency of 20 KHz and it has a deviation ratio of 10. What is the bandwidth required for transmission.

Solution:

$$D.R = (f_H - f_c)/f_m = 10 \Rightarrow f_H - f_c = 10 f_m = 200\ kHz.$$

Thus we need a total of 400 KHz of bandwidth (200 KHz on each side of the carrier frequency).

EXAMPLE: JEE MAIN: Modulation frequency of an AM radio station is 250 kHz which is 10% of the carrier wave. If another AM station approaches you for a licence, what broadcast frequency will you allot?

(a) 2750 KHz (b) 2900 KHz (c) 2250 KHz (d) 2000 KHz.

Solution:

$f_m = 10\% \times f_c \Rightarrow f_c = 2500\ kHz$. Thus this radio station occupies 2500 ± 250 i.e 2250 to 2750 kHz. Now for the second radio station also, modulating frequency will be 250 KHz. Thus its central frequency should be atleast 250 KHz away from sidebands of the first station. Accordingly, its central frequency can't lie between 2000 KHz and 3000 KHz. Thus only

YOUTUBE.COM/@THEPHYSICSSHELTER

possibility is 2000 KHz. (Bonduline! :)).

EXAMPLE: JEE MAIN: In a communication system operating at wavelength 800 nm, only 1% of source frequency is available as signal bandwidth. The number of channels accomodated for transmitting TV signals of bandwidth 6 MHz is:
(a) 3.75×10^6 (b) 4.87×10^5 (c) 6.25×10^5 (d) 3.86×10^6

Solution: Here we need to interpret the language intelligently. (Because the language is actually fairly opaque). Here possibly there are several channels, each with operating frequency corresponding to nearly a wavelength of 800 nm and 1% of the carrier frequency is to be distributed among all the channels. Going with that interpretation:

$f_c = 3 \times 10^8 / (800 \times 10^{-9})$ Hz $= 3.75 \times 10^{14}$ Hz $= 3.75 \times 10^8$ MHz.

Thus total available bandwidth for all channels: $1\% \times 3.75 \times 10^8$ Hz i.e 3.75×10^6 Hz. Now number of channels accomodated:
N = Total Bandwidth / Bandwidth per channel i.e $3.75 \times 10^6 / 6 = 6.25 \times 10^5$
Thus correct option is (c).

YOUTUBE.COM/@THEPHYSICSSHELTER